ANUBHAV KUMAR

Python Software Development | Application Development | APIs | SQL | Automation
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SUMMARY

2026 Electronics and Communication graduate who builds and tests Python applications: a FastAPI and PostgreSQL web app, and a data pipeline with model evaluation on public CDC survey data. Three-month full stack development internship working in Java Spring Boot. Seeking an entry-level Python developer or software engineer role.

EDUCATION

Bachelor of Engineering (B.E.), Electronics and Communication Engineering | Sep 2022 – Aug 2026
Nitte Meenakshi Institute of Technology (NMIT), Bengaluru

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, HTML, CSS
Python: FastAPI, Pydantic, SQLAlchemy, pytest, pandas, NumPy, scikit-learn, Streamlit
Backend & APIs: REST API design, JWT authentication, request validation, database transactions, unit and integration testing; Java Spring Boot (internship)
Databases: PostgreSQL (schema design, indexes, query-plan analysis); SQL joins, aggregation, CTEs, window functions
Frontend: React and strict TypeScript, Recharts (for the SIP dashboard)
Tools: Git, GitHub, VS Code

PROJECTS

SIP Friction Analyzer: Investment Simulator | Python, FastAPI, PostgreSQL, React, TypeScript | GitHub

- A web app that shows what skipping, reducing or pausing a monthly investment plan (SIP) costs compared with sticking to the plan: you enter a plan and the missed or reduced months, and it reports the final value, the value lost to missed compounding and a 0–100 discipline score.
- Built the backend in Python with FastAPI: it validates each request, runs the calculation and saves the run and its events to PostgreSQL in one transaction, so a run is never stored without its events. Users sign in with JWT tokens and see only their own runs.
- Wrote the calculation engine and a Monte Carlo mode that runs 1,000 random return paths and reports the 10th, 50th and 90th percentile outcomes. Checked by hand: 5,000 a month at 12% for a year with month 2 skipped loses 5,578 of final value (discipline score 91.4), matching the app.
- Designed the PostgreSQL tables (users, runs, events) with foreign keys, cascading deletes and value checks, and added an index for the per-user history lookup. On 20,000 generated rows the isolated query dropped from about 0.8 ms to 0.06 ms; the full history endpoint, which also joins events, went from about 3.8 ms to 2.8 ms (roughly 1.3 to 1.4 times faster).
- Removed a second copy of the calculation from the React and TypeScript frontend, so one backend implementation decides every result; covered sign-in, saving and cascading deletes with pytest.

Dietbot: Calorie-Intake Model and Meal Planner | Python, pandas, scikit-learn, Streamlit, PostgreSQL | GitHub

- A Streamlit app that turns height, weight, age, sport and goal into a daily calorie target, macro targets and three meal plans from a 145-food table, and can save plans in PostgreSQL. I rebuilt it from an inherited notebook prototype into a tested Python package and fixed two bugs in the inherited code.
- Wrote a script that downloads three public CDC survey files (NHANES 2017–2018), joins them with pandas and keeps adults with plausible intake, giving a clean dataset of 4,624 people.
- Found that the prototype's training data was effectively a formula (every macro value a fixed multiple of calories, correlation above 0.999 over 99 rows), so its model could only relearn that formula; replaced it with a model trained on the survey data.
- Compared a Random Forest with simple baselines on held-out people (average error in kcal): plain average 684, Mifflin-St Jeor calorie formula 651, linear regression 632, Random Forest 633. The forest was no better than linear regression and R^2 is only 0.13, so I document it as a modest estimate.

EXPERIENCE

Full Stack Development Intern | Redevelopers Automate Solution Pvt. Ltd. | Feb 2026 – Apr 2026

- Developed RESTful backend APIs in Java Spring Boot.
- Designed and managed relational database components using SQL and NoSQL technologies for application data persistence.
- Identified and resolved software defects; wrote unit and integration tests to improve application reliability.

CERTIFICATIONS

Kaggle: Intro to SQL and Advanced SQL | 2026
Career Essentials in Data Analysis | Microsoft & LinkedIn (Power BI, Python, Excel, PostgreSQL) | 2026

LEADERSHIP

- Design Team Lead, ASE Club: organised I-Solve, a national-level hackathon with Bangalore Traffic Police (100+ participants).